↳ The main differences between permutations & Combinations:-

↳ In permutations order is very important and place an important role.

↳ In combinations it doesnot matter

↳ In permutations arranging the objects with conditi-ons based.

Ex:- assigning seats, assigning places to a particular letters/numbers, 1st place (or) 2nd place ---etc on running race, ranks on subjects to particular students.

↳ In combinations: choosing the/selection of the members/letters/committee based on the require-ment on the questions.

↳ Always No. of permutations > No. of arrangements.

Ex:- How many ways we can arrange 2 members on 2 places from 4 members (A, B, C, D) (or) How many pairs we can form from 4 members (A, B, C & D) with order of selection important? [Permutations]

AB, BA, AC, CA, AD, DA, BC, CB, BD, DB, CD, DC = 12 ways

$$\Rightarrow {}^4P_2 = {}^nP_r = \frac{4!}{2!} = 4 \times 3 = 12 \text{ ways.}$$

Ex:- How many pairs can be selected from 4 members (A, B, C, D) with order of selection doesnot matter? [Combination].

AB, AC, AD, BC, BD, CD (or) BA, CA, DA, CB, DB, DC

= 6 ways

$nC_r = {}^4C_2 = \frac{4!}{(4-2)!\cdot 2!} = \frac{4\times 3\times 2\times 1}{2\times 2} = 6$ ways

$\boxed{{}^4P_2 > {}^4C_2}$ Permutations > Combinations.

↳ Circular Permutations:-

↳ The main difference b/w linear permutation / Permutation & circular permutation is in linear there is a start & end but in circular there is no start & no end particularly based on assumption we have to consider.

↳ Circular arrangement nothing but one person fixed in linear & remaining arranged in remaining positions.

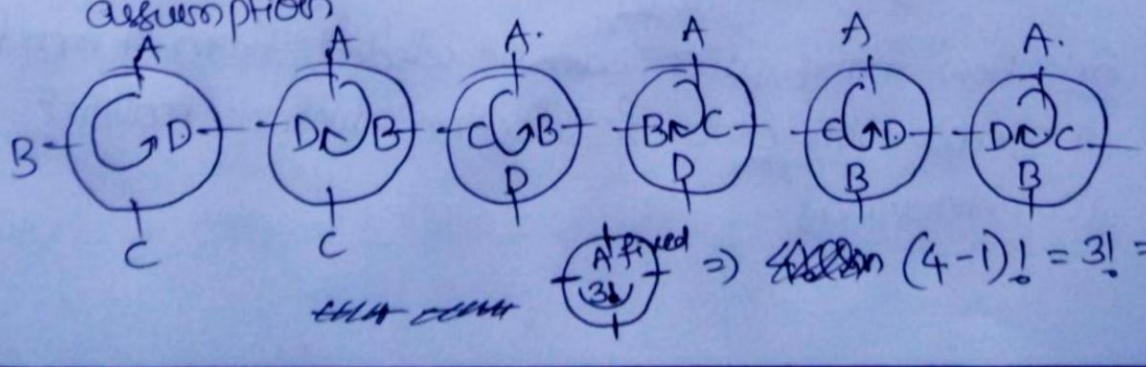
Ex:- How many ways we can arrange 4 members A, B, C & D in 4 places?

(i) linear Arrangement:- A, B, C, D

$\longrightarrow \quad \underline{4} \quad \underline{3} \quad \underline{2} \quad \underline{1} = 4! = {}^4P_4 = 24$ ways

(ii) Circular arrangement:- A, B, C & D, A is fixed

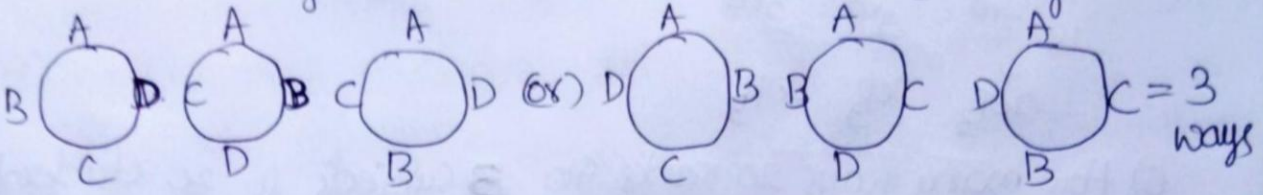
assumption



$\Rightarrow (4-1)! = 3! = 6$

↳ In the above example arrangement of objects is considered in clockwise as well as anti-clockwise.

↳ If clockwise & anti-clockwise both are same then total arrangements of circular are 3 only.



$= 3$ ways only $= \frac{(4-1)!}{2} = \frac{1}{2} 3! = \frac{3}{2}$

↳ This case is to be considered. to arrange beads to a necklace.

Example Questions:-

1) How many ways we can arrange A, B, C & D in 3 places (or) How many 3 lettered words can be formed with A, B, C & D?

$\frac{}{4}\ \frac{}{3}\ \frac{}{2} = 4 \times 3 \times 2 = 24$ (or) $4P_3 = \frac{4!}{1!} = 24$ ways.

2) How many ways 3 ranks in a subject to 20 students?

Ans:- $\frac{1^{st}}{20}\ \frac{2^{nd}}{19}\ \frac{3^{rd}}{18} = 20 \times 19 \times 18 = {}^{20}P_3 = \frac{20!}{17!} = 6840$

$= 6840$

3) How many ways 1st rank in 3 subjects to 20 students?

Ans:- $\frac{M-1^{st}}{20}\ \frac{P-1^{st}}{20}\ \frac{C-1^{st}}{20} = 20^3 = n^r$ (with repetition) $= 8000$ ways.

4) How many words can be formed with 'India'?

Ans:- 5 letters = 5 lettered words with 5 letters $= 5! = 120$ ways. words.